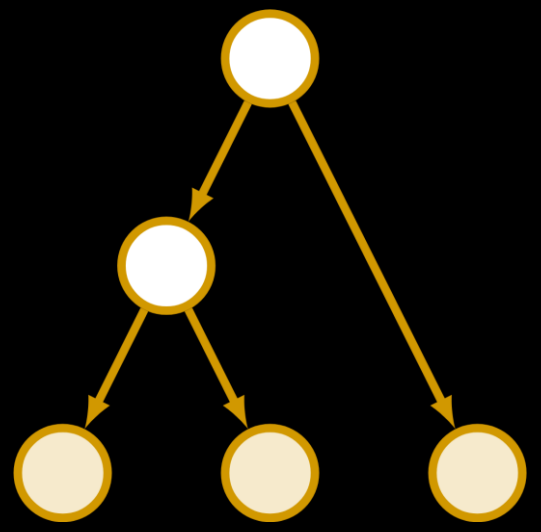


BIOL 531: RevBayes Intro

Second session: Trait Evolution, MCMC

Orlando Schwery, 13. Mar. 2026, University of Kentucky

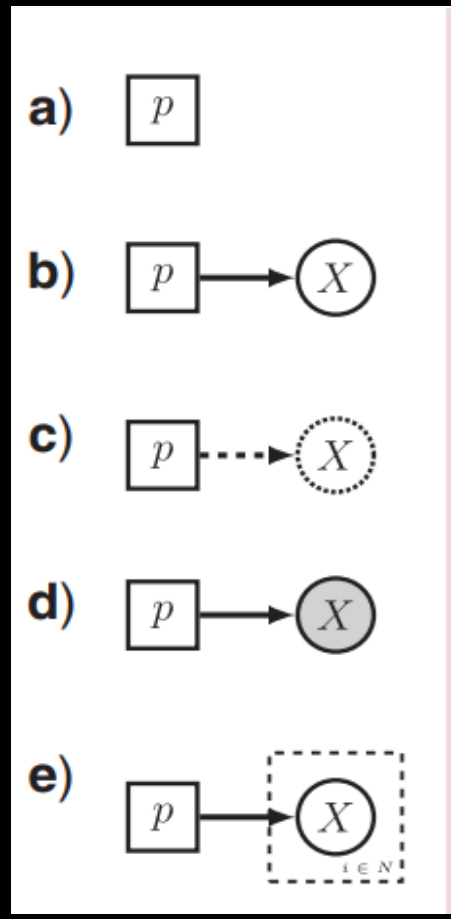
The Plan…

1. Introduction: The why and how of RevBayes
   (Basics of Rev Language and comparison to R)

2. Application: Setting up a simple model
   (Simple Brownian Motion)

3. Mix & Match: The strength of RevBayes
   (Joint inference)

Questions from Last Time?



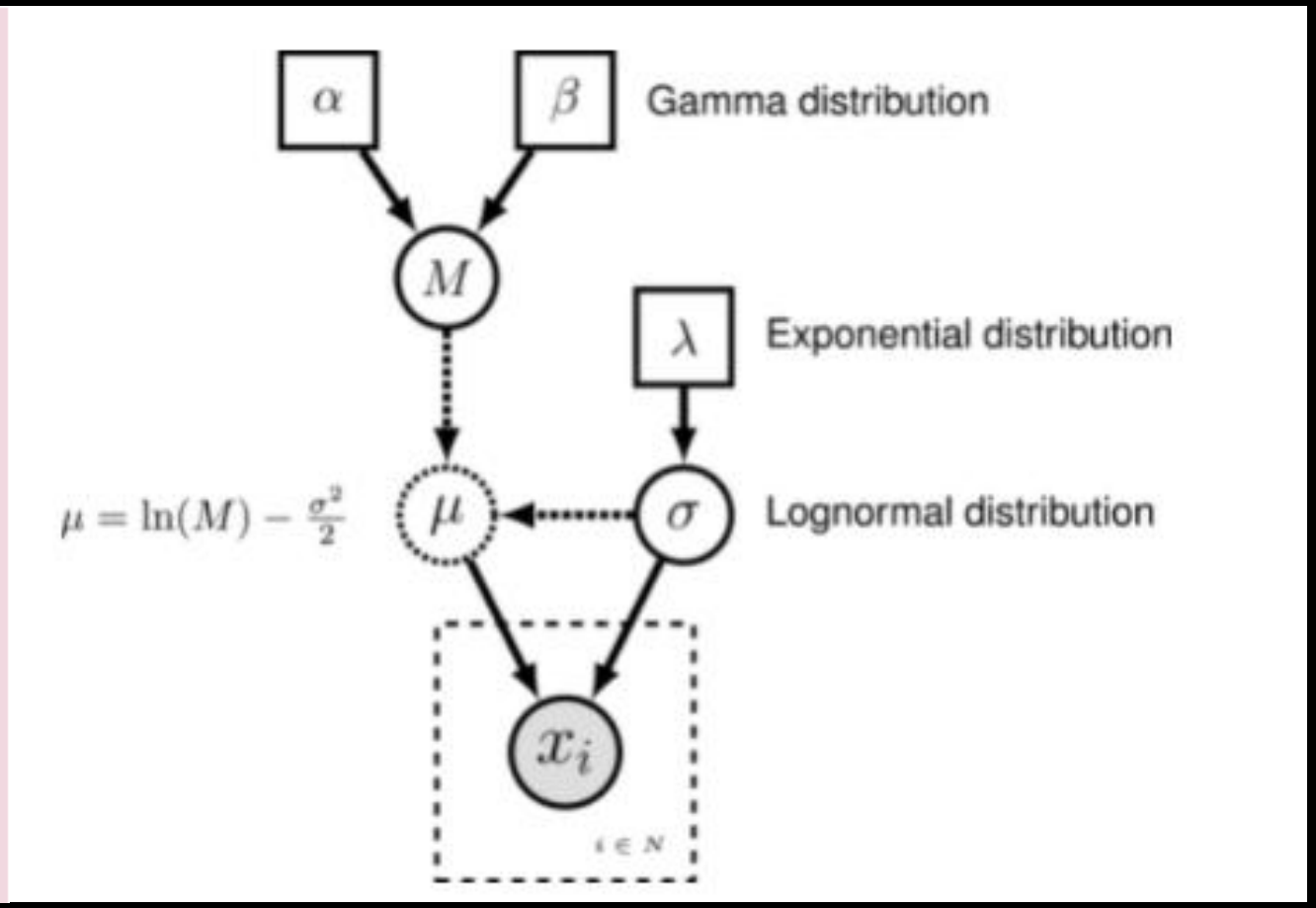

```
p <- 0.5

x ~ dnBernoulli(p)

x := exp(p)

x.clamp(1)

for (i in 1:N) {
   x[i] ~ dnBernoulli(p)
}
```



Directed Acyclic Graph (DAG)

→ Nodes and Edges

Lognormal Model/Distribution:

$X = e^{\mu+\sigma z}$ [μ: location parameter; σ: standard deviation]

Simple Brownian-Motion (Reminder)

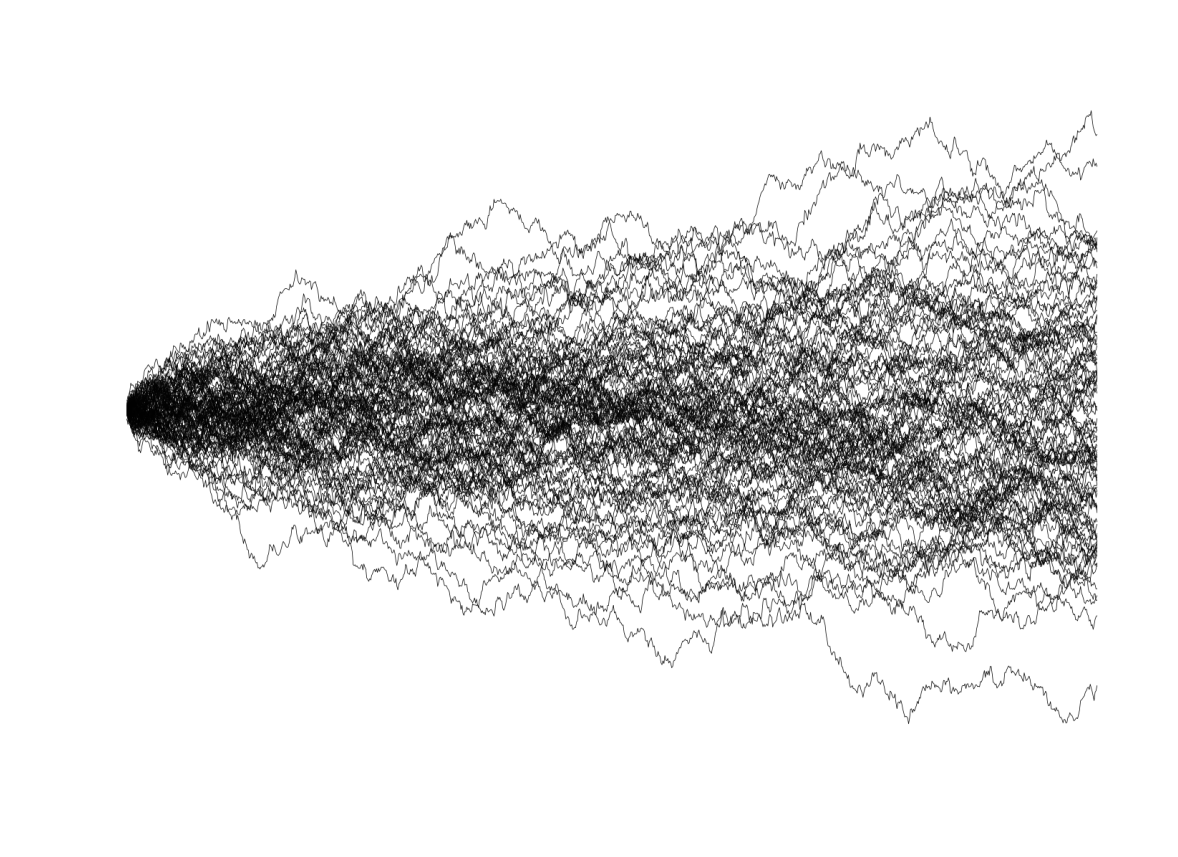
Brownian motion is a random walk

Three facts describe Brownian motion

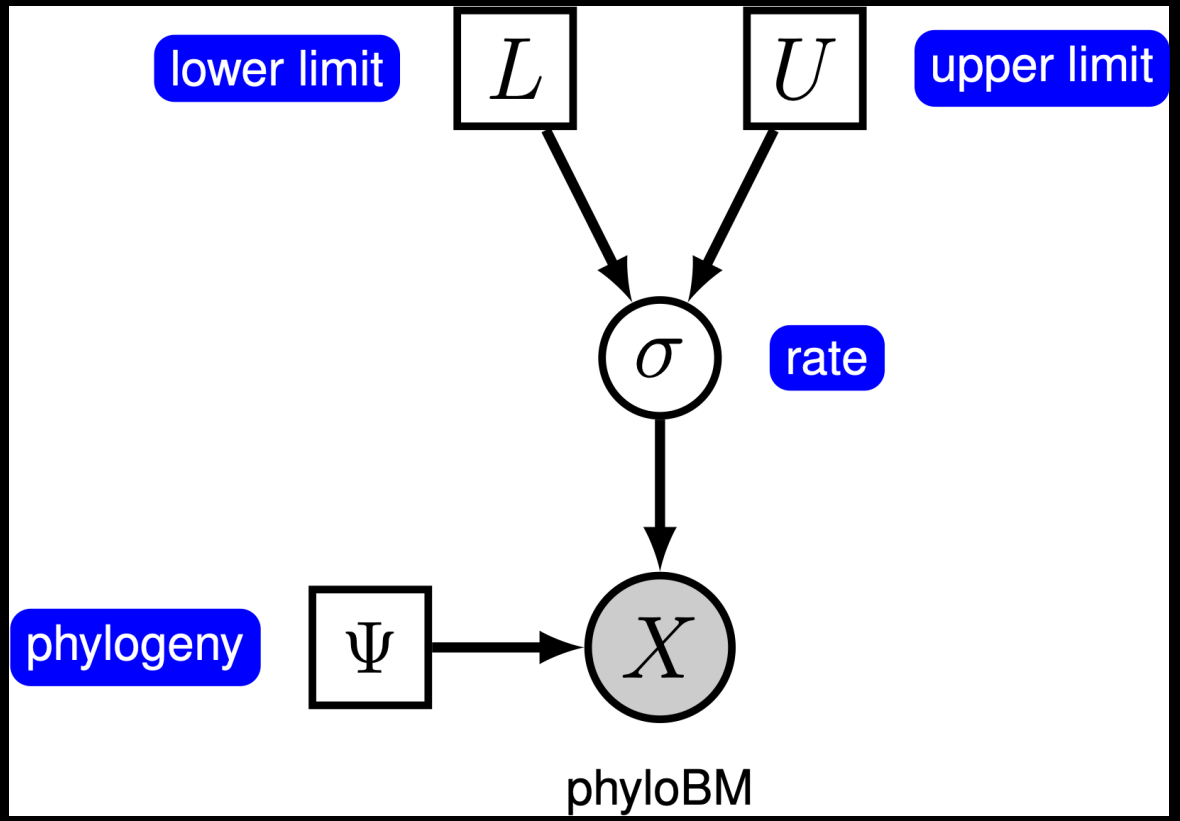
1. No trend
2. Successive steps independent
3. Expected values follow normal distribution where variance depends on rate and time

Brownian motion models have two parameters:

- x_0, the starting value
- σ^2, the rate parameter

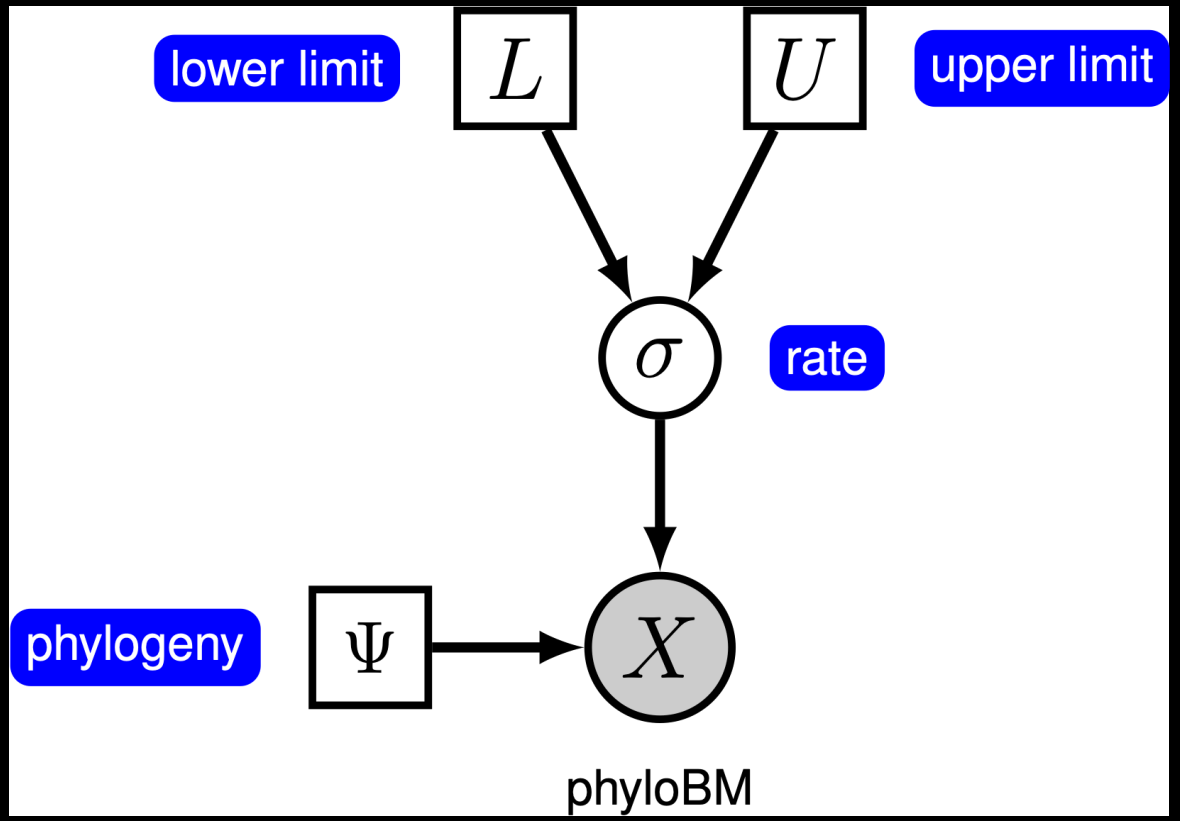


Simple Brownian-Motion (RevBayes-Style)



Simple Brownian-Motion (RevBayes-Style)

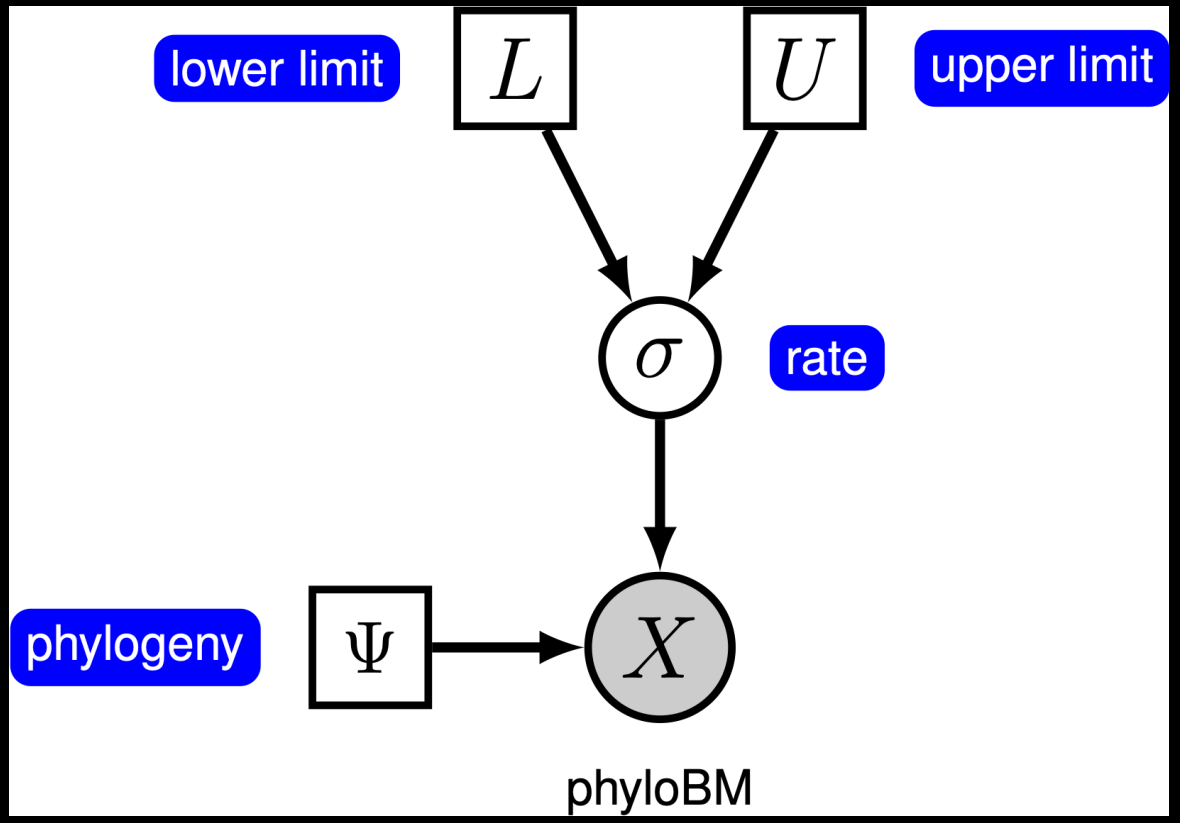
- The node X contains the BM model for trait data based on tree and rate, with observed data clamped to it



Simple Brownian-Motion (RevBayes-Style)

- The node X contains the BM model for trait data based on tree and rate, with observed data clamped to it

- Single rate parameter sigma^2
  - Drawn from loguniform distribution
  - Lower bound L
  - Upper bound U

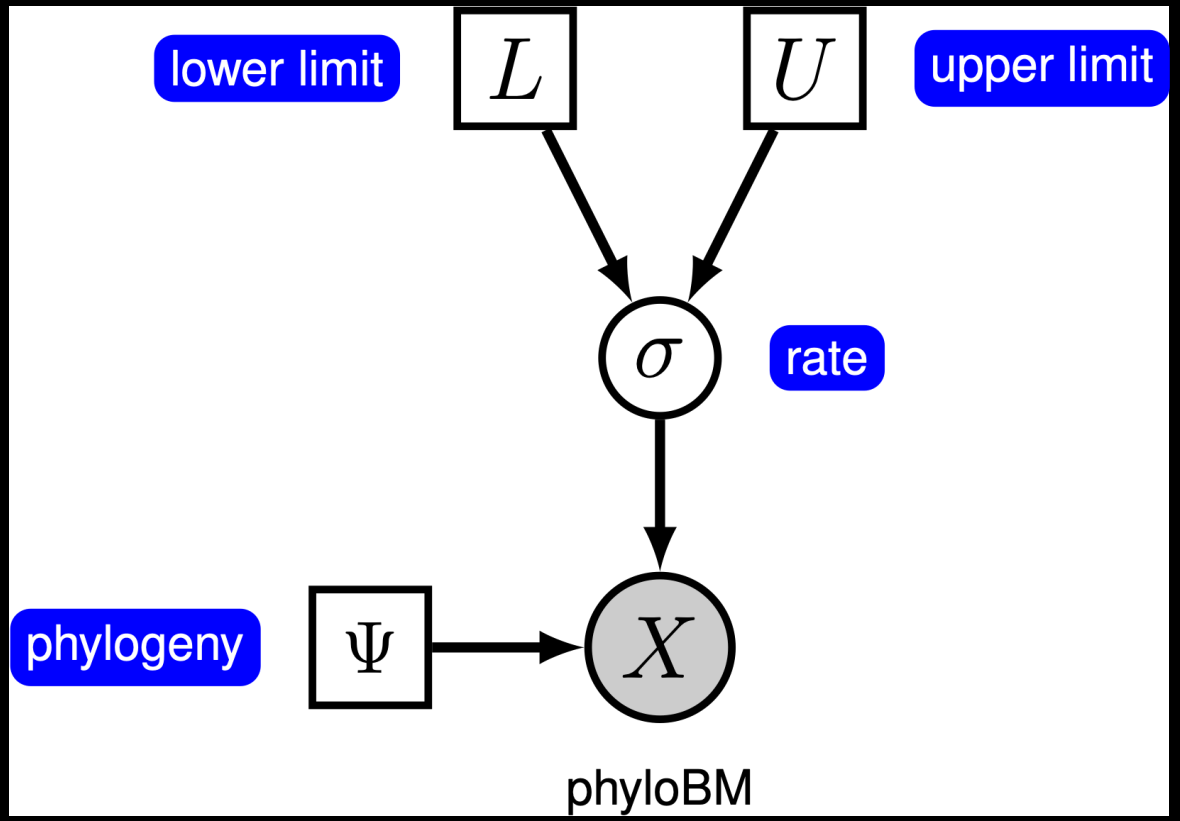


Simple Brownian-Motion (RevBayes-Style)

- The node X contains the BM model for trait data based on tree and rate, with observed data clamped to it

- Single rate parameter sigma^2
  - Drawn from loguniform distribution
  - Lower bound L
  - Upper bound U

- The phylogeny is assumed to be fixed, thus added as a constant node



Running an MCMC

- The other tutorials are doing a pretty good job at looking more in-depth at the inner workings of the MCMC, different options etc.
- We'll use it out of the box, focusing on the 'how to run' for now.
- Background:
  - Bayesian analyses: we're interested in the posterior distribution of our parameters, often can't be calculated directly, so we do it numerically using MCMC
  - 'robot on landscape' analogy
    - Move through parameter space
    - Evaluate likelihood of parameter combinations at proposed move
    - Move towards improvement to get to peak(s)